
Cell Type Image Classification

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Abstract

Cell classification is a commonly used technique in healthcare settings and biomedical research to aid in disease identification and prevention. However, established methodologies often rely on manual classification, which is time and labor intensive while also requiring a specialized skill set. As such, much work has been done in recent years to develop classification algorithms taking advantage of classical machine learning paradigms. In this project, we present an approach utilizing ResNET18 architecture to perform classification of microscopy into 14 cell line categories from the CTMC-v1 dataset. Among the OFAT experiments, all configurations except Experiment 5 showed modest performance gains over the baseline, with Experiment 1 achieving the best overall results (accuracy: 0.7720, F1 score: 0.6970, AUROC: 0.9730), while Experiment 5 showed reduced accuracy despite improvements in F1 and AUROC. Overall, our model performs well on classification for 12 out of 14 cell lines (accuracy ≥ 0.5), but struggles on male human lung carcinoma and normal rabbit kidney.

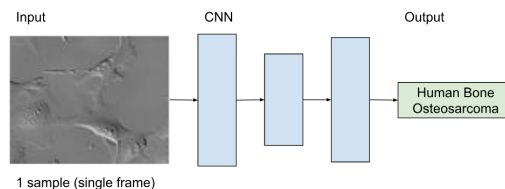


Figure 1: Simplified outline of classification model.

1 Introduction

This project is based on the CTMC-v1 Cell Tracking with Mitosis Detection Dataset Challenge, which was originally hosted at <https://motchallenge.net/data/CTMC-v1/> (the original page has gone down but it has been archived at <https://web.archive.org/web/20260304154036/https://motchallenge.net/data/CTMC-v1/>).

In modern medicine and biomedical research alike, cell classification remains a commonly used technique to aid in the characterization and diagnoses of diseases. For instance, oncologists may be concerned with cancer cell development (1), or geneticists may need to categorize morphology to understand cell development (2). However, there are limitations in traditional methods of manual classification that have become increasingly apparent with time.

In modern research, there has been an emphasis on the development of high-throughput multiplexed approaches that generate large volumes of cell imaging data to be processed (3) (4) (5). On one hand, these methodologies have drastically improved the speed and accuracy at which research

can be performed due to high volume processing of samples. However, the process also becomes bottlenecked in data analysis without the automation of classification. Traditional methods utilize human judgment for cell classification, which is unsustainable for high quantities of data. These methods are often time consuming and laborious while also requiring the practitioner to have a certain level of expertise. This makes scaling with data difficult, and can also result in increased levels of inaccuracies due to human error.

Additionally, in healthcare settings, cell classification of biopsies streamlines diagnoses and allows physicians to provide the care necessary for patient health. However, once again, this often is a time and labor intensive process, while also requiring a trained pathologist to manually sift through H&E stained biopsy slides to identify key cell types. These inefficiencies are especially problematic in conditions such as Hirschsprung’s Disease, where the presence of a single cell can determine whether a diagnosis is conclusive or not (6). Furthermore, in low and middle income countries, the capacity to take biopsies is present, but is often lacking the necessary skilled physicians to interpret them. This results in patients often not getting the care they need.

In this paper, we outline one approach to classification of cell lines using a CNN for use in high-throughput cell classification based on morphology feature extraction in DIC microscopy imaging. The implementation of such a network would optimize the classification process by moving from traditional manual classification methods that are slow and often prone to human error associated with fatigue and subjectivity. We provide an overview of the data, a theoretical framework of the CNN model we are choosing, and evaluation on model performance against defined baselines using several metrics. Our model achieved an accuracy of 0.7579, an F1 score of 0.6728, and an AUROC score of 0.9722 in the public CTMC-v1 challenge dataset.

2 Related Work

There have been many previous works involved in automating the identification and categorization of cells by neural networks. This includes categorization for oncology, development, and more.

Much has been done previously on the subject of applying deep learning to cell image classification tasks. CNNs are well-suited to classifying image data and have become the standard tool for image classification, learning patterns of visual features directly from raw pixel data through layers of convolution, which extracts features, and pooling, which downsamples feature maps to reduce dimensionality. Early work has shown that CNNs outperform traditional machine learning pipelines on a variety of biomedical imaging tasks, and have now become the standard tool for image classification.

One such work by Chuang et al. was centered on the classification of cancer types and tumor tissue. They used a combination of spectral clustering and CNN techniques to predict and classify 11 cancer types using protein interaction networks and omics data, helping them achieve an accuracy of 95.4% (7). While this work operated on omics data rather than raw microscopy images, it demonstrates the effectiveness of CNN-based approaches for multi-class biological classification. Our work differs in that we classify directly from raw DIC microscopy images rather than molecular data, addressing a larger and more visually similar set of classes.

Prior work has also demonstrated successful CNN-based classification of cell lines directly from microscopy images. Klemba et al. applied a CNN to fluorescence microscopy images of breast cell lines and successfully distinguished between three classes based on intracellular actin filament organization which would otherwise be difficult for humans to identify (8). Silveira et al. achieved 93% classification accuracy by applying CNNs to brightfield microscopy images of cell lineages (9). While both works demonstrate the viability of CNN-based cell line classification, they operate on fluorescence or brightfield microscopy with few classes. In contrast, our task involves 14 cell lines imaged under DIC microscopy, a more challenging setting with greater inter-class visual similarity.

Most relevant to our approach, Hu et al. were able to demonstrate that transfer learning with pretrained CNNs can classify cellular morphological changes from microscopy images and outperform models trained from scratch, with the added benefit that these pretrained CNNs require less training data (10). Our work builds directly on this paradigm, applying a pretrained ResNet18 backbone fine-tuned on DIC microscopy frames, extending the approach to a more complex 14-class cell line identification problem.

Conventional neural networks have also been used for the classification of microscopy images that were collected using DIC imaging. One study trained CNN models to classify the stages of embryonic cell division and was able to achieve an average F1 score of 95% across species (11). This is directly relevant to our work as it shows CNNs can effectively discriminate features from the same DIC imaging modality we use. However, that work classifies division stages within a single organism rather than distinguishing across 14 distinct cell lines, which is our more challenging objective.

The CTMC challenge paper itself benchmarked four modern tracking algorithms (Viterbi, DeepCell, Tracktor, and DeepSORT) on DIC microscopy data across 14 cell lines (12). While these algorithms were designed for tracking rather than classification, their analysis revealed important characteristics of the dataset. High-density cell lines such as BPAE, CRE-BAG2, and OK proved significantly more difficult than low-density lines such as CV-1 and A-10. Unlike the tracking approaches benchmarked in that challenge, our model addresses the problem as a frame-level classification task using transfer learning.

However, there is often disparity in data availability between different cell types in publicly available datasets, and this class imbalance presents a common challenge in biomedical image classification. Prior work has established that training CNNs on imbalanced datasets without correction results in models heavily biased towards majority classes, significantly degrading minority class performance (13). This is also apparent in the CTMC dataset, and so we attempt to minimize this by applying weighted loss functions during training and using data augmentation to increase the representation of underrepresented classes.

3 Methods

We employ transfer learning using a ResNet18 backbone pretrained on ImageNet (Fig. 1). The final fully connected layer is replaced with a custom classification head consisting of a dropout layer ($p=0.5$) followed by a linear layer mapping the 512-dimensional feature representation to 14 output classes. The model takes a single grayscale frame (replicated to 3 channels to match ResNet18 input format) and outputs a predicted cell line label.

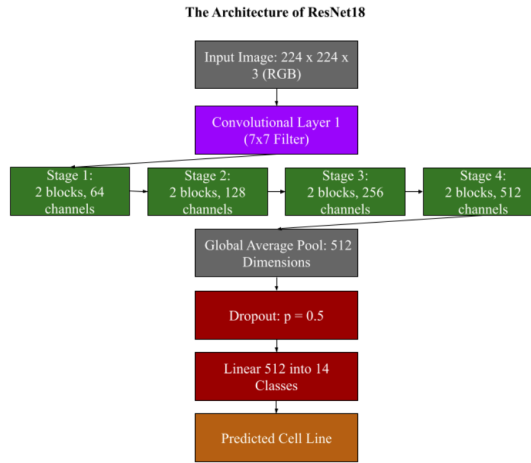


Figure 2: ResNET18 architecture for the development of our cell classification CNN

We saw class imbalance in our dataset, so we applied a weighted cross-entropy loss function that assigned higher penalties when underrepresented cell lines were misclassified, with weights inversely proportional to class frequency. We additionally apply data augmentation during training including random horizontal and vertical flips, random cropping with padding, and brightness and contrast jitter to increase effective sample diversity for rarer classes. These techniques have been shown to substantially improve minority class performance in CNN-based biomedical image classification (13).

During fine-tuning, the backbone is trained with a lower learning rate ($1e-4$) than the classification head ($1e-3$) using the Adam optimizer, allowing pretrained features to adapt gradually while the head trains more aggressively. A cosine annealing learning rate scheduler is applied over 20 epochs.

Our primary goal is to achieve a least 50% overall classification accuracy on the CTMC-v1 test frames. On a 14-class problem, this represents roughly seven times better than random chance and constitutes a meaningful result. We anticipate that high-density cell lines such as BPAE, CRE-BAG2, and MDBK will be harder to classify correctly, consistent with difficulty patterns observed in the original challenge [7], and that overall accuracy may be pulled down by these harder classes.

4 Experiments

We work with the CTMC-v1 dataset (12), which contains 86 live-cell imaging video samples representing 14 different cell lines, for a total of 152,584 frames collected using DIC microscopy at 30 fps and 320x400 resolution. We follow the original challenge split: 47 training videos (odd-numbered sequences) and 39 test videos (even-numbered sequences), yielding 80,389 training frames and 72,109 test frames. Each frame is labeled with one of 14 cell line categories. To manage computational cost, we sample every 15th frame during training.

Note that the CTMC challenge was originally designed for cell tracking (following individual cells across frames), whereas our task is cell type classification from individual frames. These are distinct problems, and consequently the tracking metrics reported in the original challenge (MOTA, TRA) are not directly comparable to our classification metrics.

In addition to our primary goal of achieving a minimum 50% accuracy, we compare our fine-tuned model against two baselines to determine the soundness of our approach. First, a majority-class classifier that always predicts the most frequent cell line regardless of input, which is approximately 12% accuracy since that was the frequency of the most common line. Second, a frozen backbone baseline in which the ResNet18 backbone weights are kept fixed at their pretrained ImageNet values and only the classification head is trained. This directly quantifies how much the fine-tuning step contributes beyond using off-the-shelf pretrained features.

In our experiments, model performance was evaluated with several key metrics: accuracy, F1 score, and AUROC score. Given the labels for each cell line in the dataset, simple accuracy was calculated as the primary metric to check model performance. Additionally, a 14 x 14 confusion matrix was generated to assess the performance of our model in each individual cell class (Figure S1). F1 score was calculated to evaluate class imbalance and balance between sensitivity and specificity using information derived from the confusion matrix. Lastly, ROC curves were visualized (Figure S2) and AUROC scores were calculated to measure the model’s ability to discriminate between classes across different decision thresholds.

Table 1: Baseline Model Parameters

Frame Sampling Rate	Every 15th
Epoch Number	5
Weight Decay Constant	$1e-4$
Dropout	0.5
Label Smoothing	0.0

Additionally, in hopes of achieving our moonshot accuracy goal of 0.9, we performed a series of 5 experiments investigating the optimization of hyperparameter tweaking on model performance using the aforementioned metrics. Baseline performance was evaluated in triplicate using parameters defined in the base model (Table 1). A one-factor-at-a-time (OFAT) approach was utilized to assess hyperparameter impact, retraining the model by modifying a single hyperparameter while maintaining all other variables at their baseline levels and assessing model metrics. Five hyperparameters were analyzed in total: frame sampling rate, epoch number, weight decay, neuron dropout percentage, and label smoothing. All experiments were performed in triplicate.

Finally, to assess the effect of cell density in classification and our model’s capacity to be density-invariant, evaluation metrics between frames categorized based on cell count were calculated and compared. Frames were categorized into three groups: low (<10 cells), medium (10-20 cells), and high (>20 cells).

Table 2: OFAT Experiment Modifications from Baseline Model

Exp. 1	Frame Sampling Rate: every 15th $\rightarrow$ every 10th
Exp. 2	Epoch Number: 5 $\rightarrow$ 10
Exp. 3	Weight Decay: $1e-4 \rightarrow 1e-2$
Exp. 4	Dropout: 0.5 $\rightarrow$ 0.1
Exp. 5	Label Smoothing: 0.0 $\rightarrow$ 0.25

4.1 Results

Per epoch accuracy and loss were determined while training the model using the hyperparameters defined in Table 1 (Figure 3). Model parameters resulting in the highest testing accuracy were saved for later comparison of our model to defined baselines. The baseline model easily achieves the minimum 0.5 accuracy target with no further augmentation required, but falls short of the 0.9 moonshot goal.

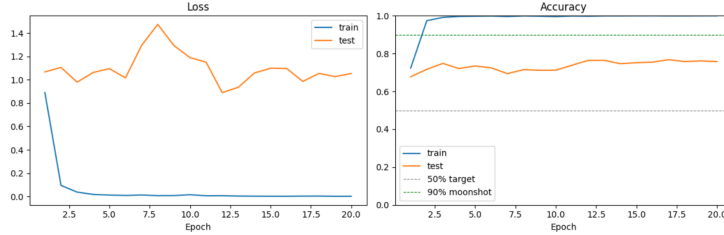


Figure 3: Per epoch loss and accuracy of model trained using baseline hyperparameters.

Quantitative macro-averaged results comparing model performance to defined baselines (Majority Class and Frozen Baseline classifiers) are shown in Table 3. Our trained model exhibits an accuracy of 0.7579, an F1 score of 0.6728, and an AUROC score of 0.9722. This is a pronounced improvement over the majority class classification accuracy at 0.1215, and the frozen baseline metrics with an accuracy of 0.6202, an F1 Score of 0.5603, and an AUROC score of 0.9255. Per class results are reported in Table S1.

Table 3: Model Comparison to Majority Class and Frozen-Baseline Classifiers

Model	Accuracy	F1 Score	AUROC Score
Majority Class	0.1215	-	-
Frozen-Baseline	0.6202	0.5603	0.9255
Fine-Tuned Model	0.7579	0.6728	0.9722

Quantitative OFAT experiment results are shown in Table 4. All experiments with the exception of experiment 5 exhibited minor improvements in all measured metrics over the baseline model. Experiment 5 demonstrated a lower accuracy, but still had improvements in the F1 score and AUROC score. Experiment 1 exhibited the largest increase in all metrics over the baseline model with average of 0.7720 in accuracy, 0.6970 in F1 score, and 0.9730 in AUROC.

5 Analysis

Cell density model evaluation results are shown in Table 5. Contrary to what was hypothesized, high density classification did not perform the worst across all evaluation metrics. While this category exhibited the lowest accuracy and AUROC score, a higher F1 score was observed compared to low density classification and the accuracy was not dramatically lower than the next highest accuracy. Low density classification exhibited the highest AUROC score of 0.9986, while medium density classification exhibited the highest accuracy and F1 score of 0.8033 and 0.8258 respectively.

Table 4: OFAT Mean Experimental Metrics (N = 3)

Method	Accuracy	F1 Score	AUROC
Baseline	0.7562 ± 0.0043	0.6644 ± 0.0052	0.9639 ± 0.0028
Exp. 1	0.7752 ± 0.0052	0.6970 ± 0.0059	0.9730 ± 0.0042
Exp. 2	0.7720 ± 0.0099	0.6902 ± 0.0179	0.9677 ± 0.0045
Exp. 3	0.7594 ± 0.0090	0.6766 ± 0.0130	0.9661 ± 0.0027
Exp. 4	0.7569 ± 0.0064	0.6723 ± 0.0098	0.9672 ± 0.0017
Exp. 5	0.7437 ± 0.0101	0.6842 ± 0.0185	0.9649 ± 0.0039

Table 5: Model Evaluation Metrics Across Cell Densities

Category	Frame #	Accuracy	F1 Score	AUROC Score
Low Density (<10 cells)	1225	0.7159	0.6127	0.9986
Medium Density (10-20 cells)	2552	0.8033	0.8258	0.9864
High Density (>20 cells)	1031	0.6954	0.7645	0.9360

6 Discussion

Based on our experiments and measured metrics, our model provides a promising baseline for morphology-based cell classification, reaching an overall accuracy of 0.7579, an F1 score of 0.6728, and an AUROC score of 0.9722. However, there is clear room for future improvements.

Based on confusion matrix analysis, our model currently struggles to properly classify 2 of the 14 cell lines in the data set (male human lung carcinoma and normal rabbit kidney) and has variable accuracy across other cell lines; some cell lines exhibit high accuracy in classification with upwards of 0.978 while others cell lines are classified correctly at much lower rates, going down to an accuracy 0.527 (Table S1).

Furthermore, in the analysis experiment to determine model variation in response to frame cell density, results are somewhat inconclusive due to lack of prominent patterns in data. Results may have also been skewed due to the higher amount of frames in the medium density category available for classification. Further experimentation is necessary with similar-sized classification datasets to better determine effect of cell density.

Lastly, an OFAT experimental design, while less computationally expensive, does not explore the hyperparameter space as well, which can fail to capture interactions between variables. Future work can employ the use of a DOE structure instead at a wider range of values, to further optimize hyperparameters to better aim for the moonshot 0.9 accuracy goal.

7 Conclusion

In this report, we have defined a CNN with a ResNET18 backbone trained on ImageNet for the high-throughput classification of cells based on morphological features. The model was evaluated against a majority-class and frozen-backbone classifier as baselines (in addition to a defined 50% minimum accuracy) using accuracy, F1 score, and AUROC score as performance metrics. The model exhibited better performance on all three metrics compared to these baselines. Additionally, to assess model variability for cell classification in response to cell density, these performance metrics were calculated and compared across frames with varying cell counts. Further experimentation is necessary to make a conclusion on model variability with cell density. Finally, to evaluate the impact of hyperparameters on model performance and to aim for the moonshot 90% accuracy goal, an OFAT experimental design was done in triplicate examining 5 different hyperparameters. This experiment found rate of frame sampling to have the greatest impact on model performance.

8 Supplemental Figures

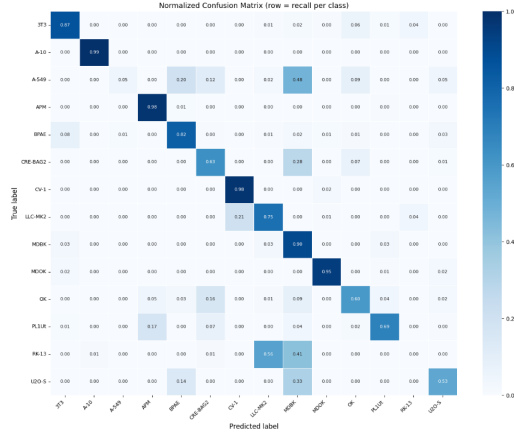


Figure S1: Normalized per-class confusion matrix for model accuracy assessment.

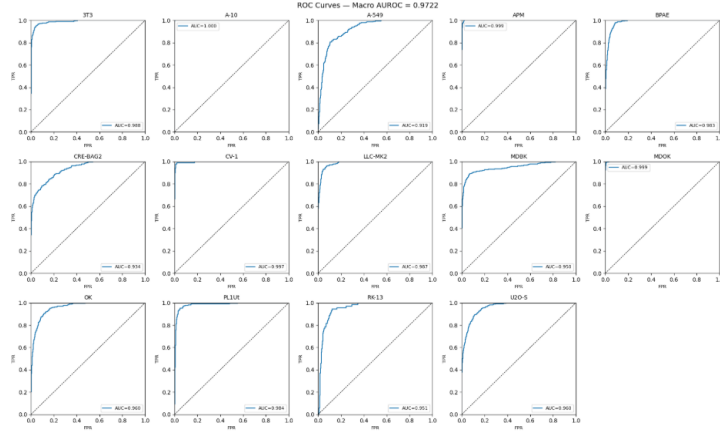


Figure S2: Visualized ROC curves for AUROC metric calculation across all 14 cell lines.

Table S1: Classification performance metrics for each cell line.

Label	Name	Accuracy	F1-Score	AUROC
3T3	Albino Swiss Mouse Embryo	0.865	0.866	0.988
A-10	Embryonic Rat Thoracic Aorta Medial Layer	0.994	0.994	1.000
A-549	Male Human Lung Carcinoma	0.047	0.088	0.919
APM	African Water Mongoose Skin	0.979	0.930	0.999
BPAE	Bovine Pulmonary Artery	0.818	0.762	0.983
CRE-BAG2	Albino Swiss Mouse Embryo (Moloney Murine Leukemia Virus Transfected)	0.632	0.633	0.934
CV-1	Normal African Green Monkey Kidney	0.978	0.734	0.997
LLC-MK2	Rhesus Monkey Kidney	0.751	0.748	0.987
MDBK	Madin-Darby Bovine Kidney	0.904	0.679	0.950
MDOK	Madin-Darby Ovine Kidney	0.946	0.963	0.999
OK	Opossum Kidney Cortex Proximal Tubule	0.598	0.647	0.960
PL1Ut	Raccoon Uterus	0.687	0.734	0.984
RK-13	Normal Rabbit Kidney	0.000	0.000	0.951
U2O-S	Human Bone Osteosarcoma	0.527	0.641	0.960

Acknowledgments

Large language model tools were used for coding assistance in this project, such as in the Google Colab code.

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